

SAI POOJITHA SAJJAVARAPU

Backend Engineer

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Education

Rajiv Gandhi University of Knowledge and Technologies (IIIT Nuzvid)

Bachelor of Technology in Computer Science, **CGPA: 8.5**

Nuzvid, Andhra Pradesh

September 2022 – April 2026

Experience

Quantum Gandiva AI

Backend Engineer

AI Backend and Data Engineer Intern

Visakhapatnam, India

Jun 2026 – Present

Nov 2025 – May 2026

- Own the **backend architecture** of a production **multi-agent AI platform** for a **UK enterprise client**, from system design through deployment, orchestrating **8+** services and multi-tool agent workflows.
- Design and maintain an **event-driven architecture** on **Redis Streams** with consumer groups for asynchronous inter-service messaging and agent coordination, decoupling services so each scales and fails independently.
- Built **third-party app integrations** and **MCP (Model Context Protocol)** server integrations so agents act inside external tools through one standard interface, handling auth, rate limits, and upstream failures.
- Build and tune **FastAPI microservices** with async request handling and service-level optimizations, cutting API latency by **~40%** under production workloads.
- Run the services on **AWS EC2, S3, Lambda, and RDS** with **Docker** and **CI/CD** on GitHub Actions, and own production operations end to end: deployments, monitoring, debugging, and resolution of live incidents.

Parabola9

AI/ML Intern

Nuzvid, Andhra Pradesh (Onsite)

December 2024 – May 2025

- Trained and evaluated **LLM-based and generative-AI models**, owning data preparation, model integration, and evaluation; **improved task accuracy** through systematic tuning and cleaner training data.
- Optimized model inference through **performance tuning** and efficient serving, reducing response time on target workloads.
- Partnered with engineers and domain experts to **integrate models end to end** into application workflows.

Projects

MediBuddy – Medical-AI Web Application 🏥

- Built a **full-stack medical AI application** that analyzes medicine images using the **Gemini Vision API** and generates structured reports with medicine details, usage, dosage, side effects, and warnings.
- Developed the **FastAPI backend** with SQLAlchemy and PostgreSQL for user accounts and analysis history, with **JWT authentication** and bcrypt password hashing for secured APIs.
- Implemented **voice-based interaction** to let users access medicine information through spoken input and audio responses, alongside the image-based analysis workflow.
- **Tech Stack:** React, FastAPI, SQLAlchemy, PostgreSQL, JWT, Docker, Gemini Vision API

SignLang – Sign Language Translation App 🗣️

- Built a sign language translation application with a **Flutter client** and **Django REST Framework** backend, using a **TensorFlow CNN** to convert sign images into text.
- Implemented the complete **image preprocessing and inference pipeline**, including a cached model loader to avoid loading the trained model for every API request.
- Added **text-to-fingerspelling** functionality and stored user translation interactions in the database, with API validation and error handling.
- **Tech Stack:** Django REST Framework, TensorFlow, SQLite, Python, Flutter

Technical Skills

- **Programming Languages:** Java, C, C++, Python, JavaScript, Dart
- **Technologies & Frameworks:** Node.js, FastAPI, Django, REST APIs, **Agentic AI**, LLMs, **Generative AI**, NLP, Machine Learning, Deep Learning, Flutter, ReactJS
- **Tools & DevOps:** Git, Docker, Postman, GitHub Actions, CI/CD, VS Code, **Redis (Streams)**, Kubernetes, Terraform
- **Databases & Cloud Platforms:** MySQL, PostgreSQL, AWS (EC2, S3, Lambda, RDS, VPC), GCP, Azure

Achievements

- **Smart India Hackathon 2024 – National Winner (ML Mentor):** Mentored the six-member team that won **1st Prize** at India's largest hackathon on **bus bunching** in Delhi bus services; built the **Conv1D + BiLSTM + Attention delay-prediction model** (Keras Tuner tuned) that the Flask and Kafka backend served to the driver app.
- **Teczite Mega Expo 2025 – 1st Prize :** Built an **AI-powered security surveillance system** combining gesture recognition, YOLO-based anomaly detection, and real-time video captioning.
- **IIIT Hackathon 2024 – 1st Prize (Parabola9):** Won for **VideoGPT**, a video captioning tool; built the video-analysis module running **InternVL2** inference with frame sampling and dynamic aspect-ratio tiling after redundant-frame elimination. The win led to the Parabola9 internship.
- **Problem Solving:** **535 LeetCode problems** solved (83 hard), contest rating **1,703** (peak 1,767, **top 14%** globally).